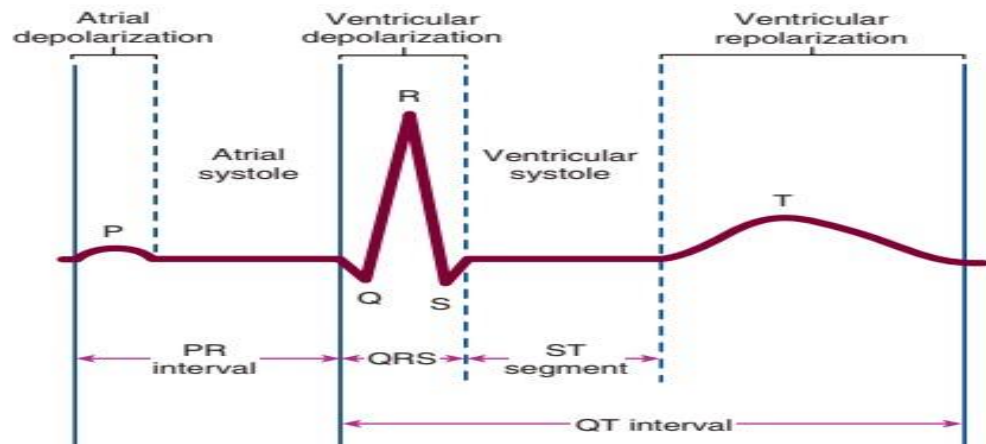
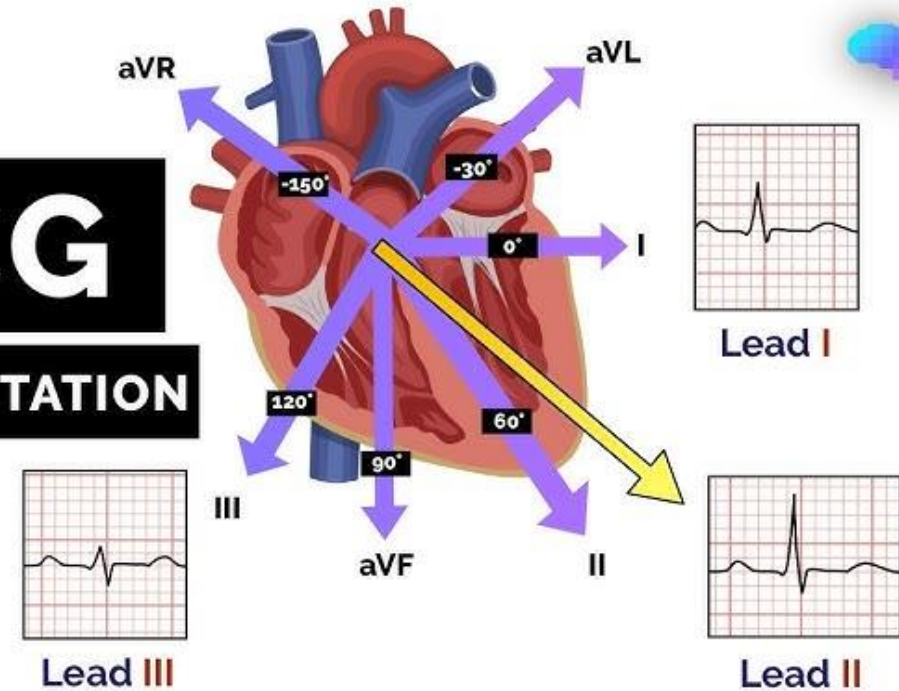


Electrocardiogram Interpretation (ECG)



ECG INTERPRETATION



Prepared by
Dr Ghada Elsayed

OUTLINE:

- Introduction
- Cardiac conduction system
- Waveforms and Intervals
- Analyzing a Rhythm Strip
- NORMAL SINUS RHYTHM

pre test

- what's the meaning of ECG
- Mention how to count heart rate for sinus rhythm

Introduction

- The ECG detects a summation of electrical signals generated by specialized
- cells of the heart called pacemaker cells.
- Pacemaker cells have the property of automaticity meaning these cells can generate a stimulus or an action potential without outside stimulation.
- This electrical signal is conducted through specialized fibers of conduction system to mechanical or muscle cells of the heart where a cardiac contraction is generated

Cardiac conduction system:

- -SA node (Sinoatrial) node
- -AV node (Atrioventricular) node
- -Bundle of His
- -Purkinje Fibers

Depolarization.

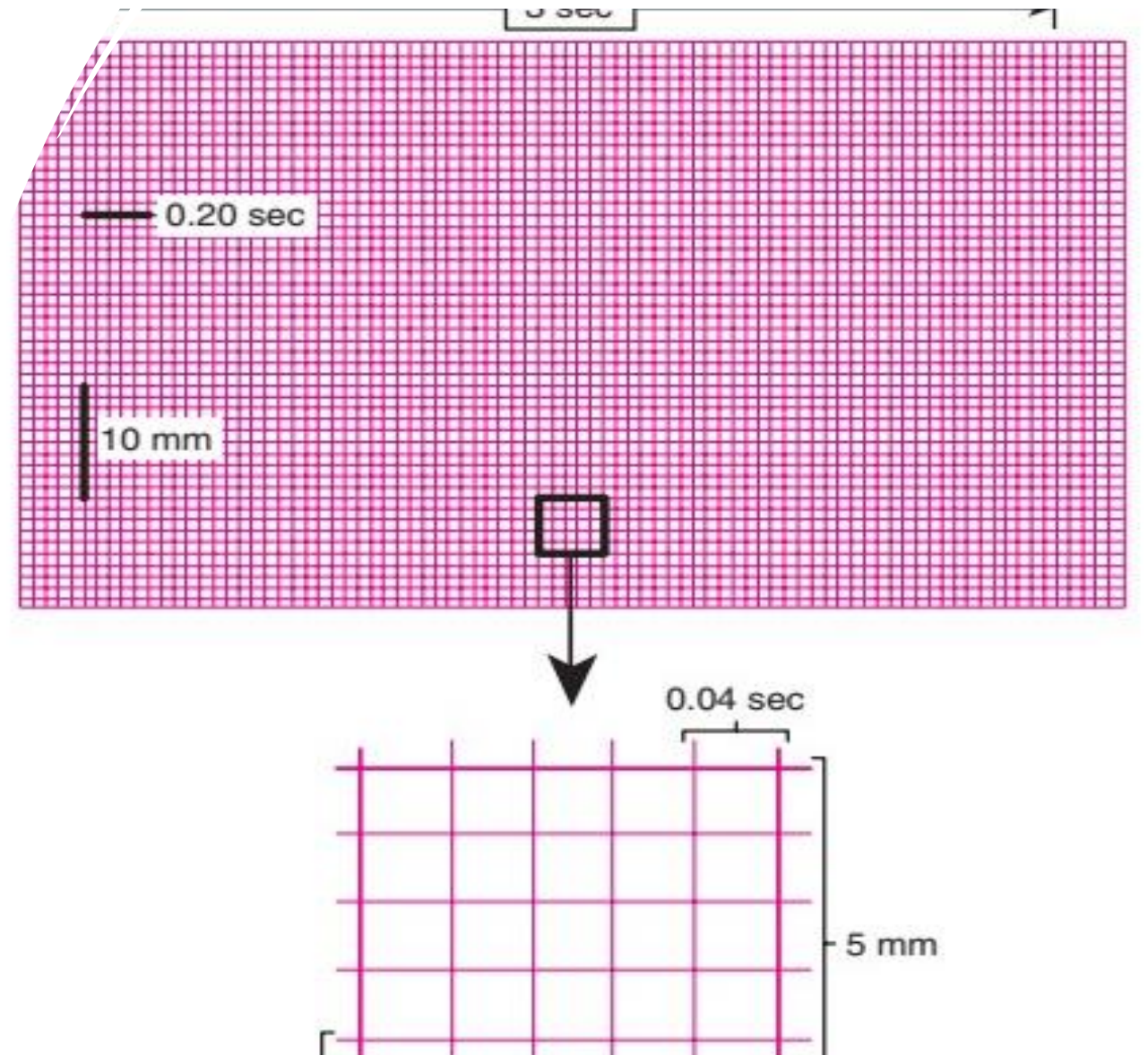
- Following these, the plateau phase occurs when calcium channels open and calcium flows into the cell. This slower phase allows for a longer period of depolarization.
- *The next event of an action potential occurs when the cell returns to its resting state.
- These are called **repolarization**.
- Calcium and Sodium are returning outside the cell, where the Potassium is inside the cell.

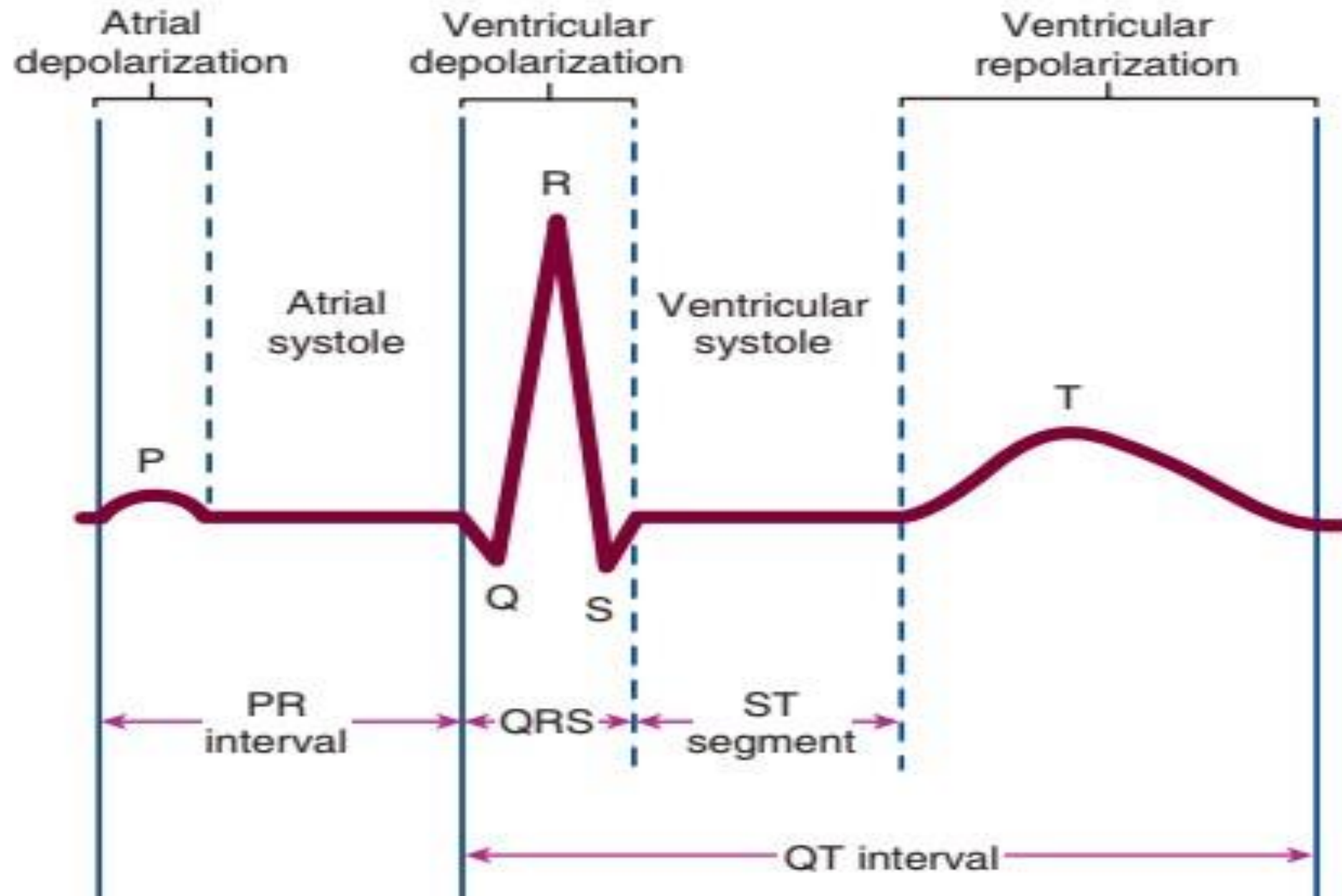
Waveforms and Intervals

- ECG paper records the speed and magnitude of electrical impulses on a grid composed of small and large boxes. Every large box has five small boxes in it.
- At a standard paper speed of 25 mm/sec, on the horizontal axis, one small box (1 mm) is equivalent to 0.04 second, and one large box (5 mm) represents 0.20 second.

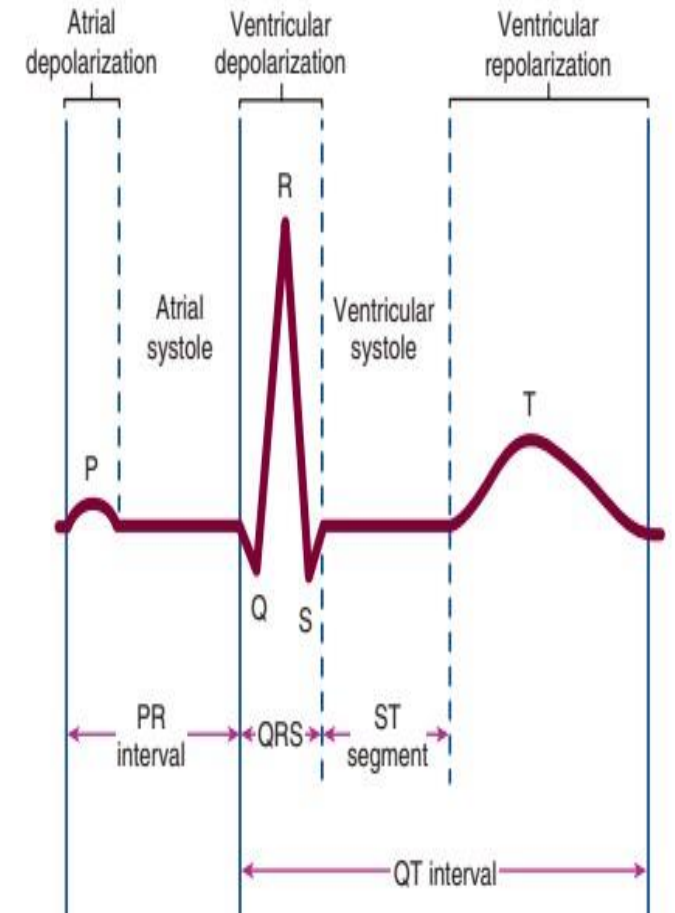
During the cardiac cycle, the following waveforms and intervals are produced on the ECG surface tracing.

- P, ORS, and T wave, these waveforms rise from a flat base line called isoelectric line.

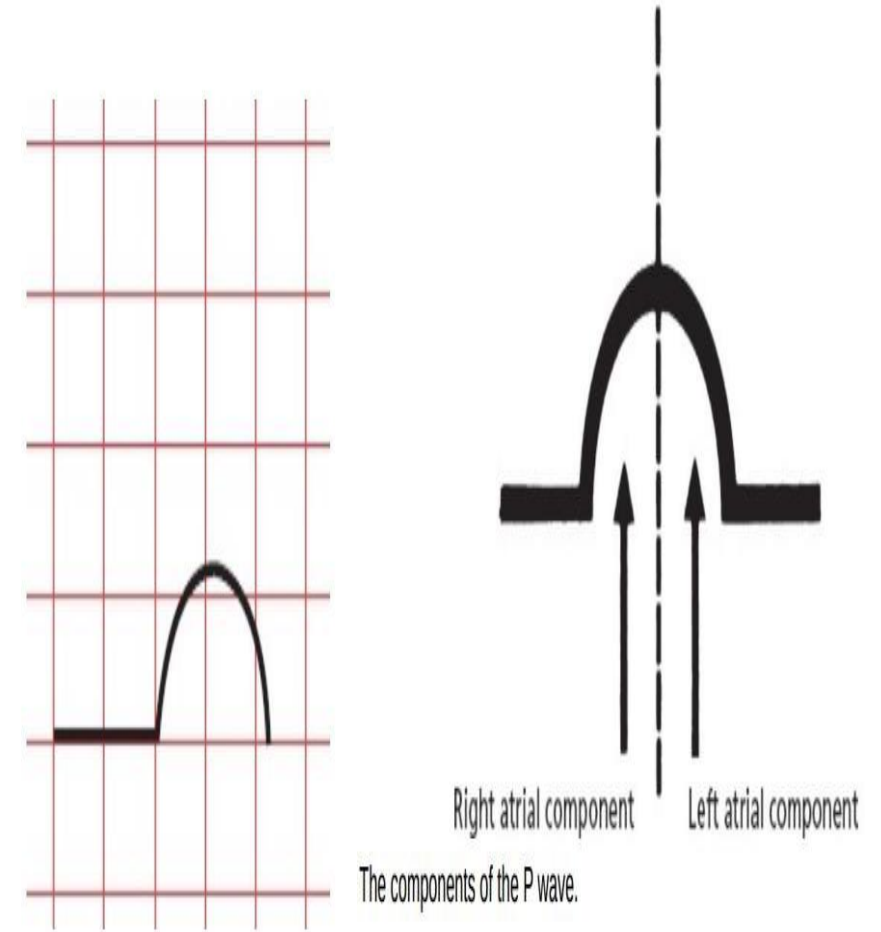




- The **P wave** represents atrial depolarization, followed immediately by atrial systole.
- The **QRS represents** ventricular depolarization, followed immediately by ventricular systole.
- The **ST segment** corresponds to phase 2 of the action potential, during which time the heart muscle is completely depolarized and contraction normally occurs.
- The **T wave** represents ventricular repolarization.
- The **PR interval**, measured from the beginning of the P wave to the beginning of the QRS,

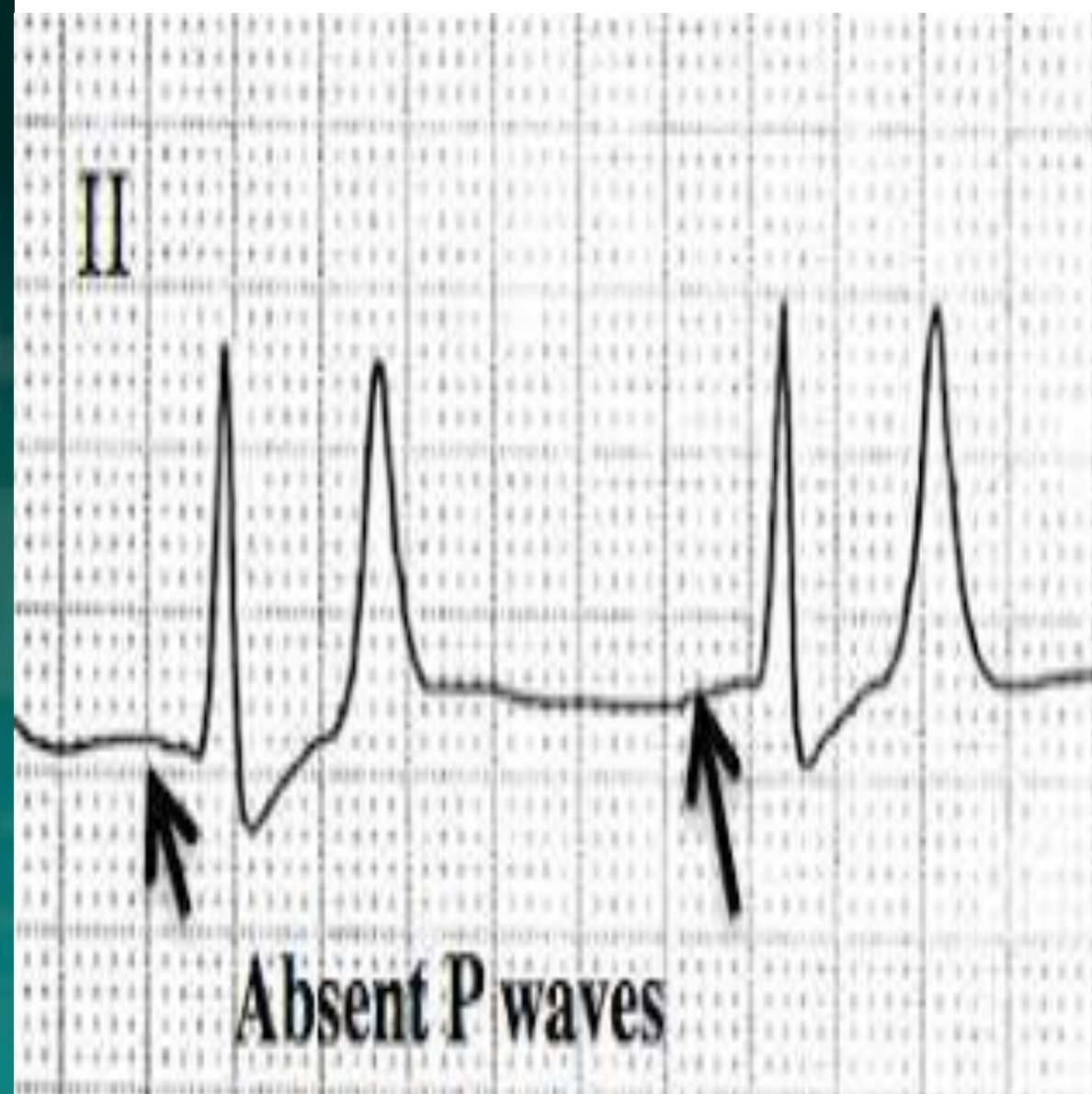


- **P wave**
- Characteristic of P wave
- ***Location:** It normally seen before the QRS complex at a consistent interval.
- ***Duration:** From 0.06 to less than 0.12 second.
- ***Configuration:** The P wave is a small, usually upright and rounded deflection, representing depolarization of the atria.



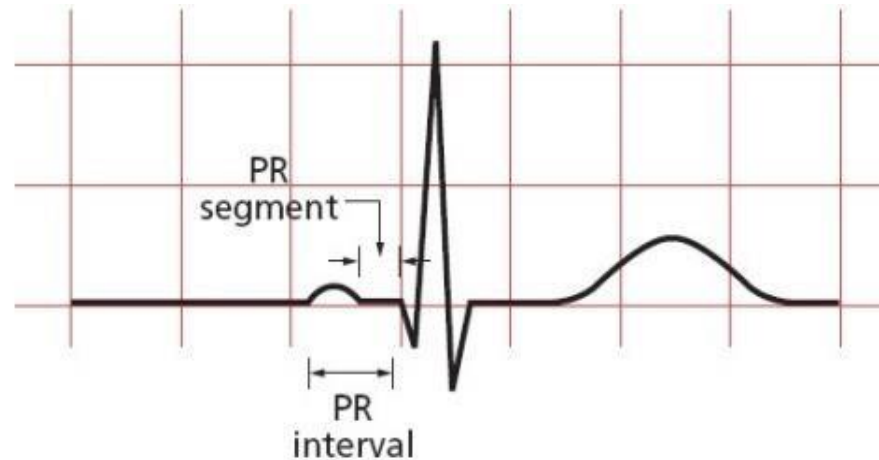
Left Atrial Enlargement

- Notched/bifid ('M' shaped) P wave (P 'mitrale') in limb leads



PR Interval

- The PR interval represents the time from the onset of atrial depolarization until the onset of ventricular depolarization.
- The PR interval is measured from the beginning of the P wave to the beginning of the QRS complex. A normal PR interval is 0.12 to 0.20 second.

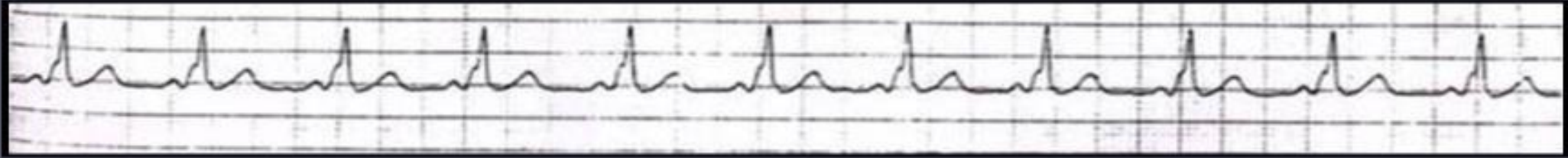


First-Degree AV Block



- Abnormal PR interval
- Prolonged PR Interval

WPW



- **Delta wave, short PR interval**

- Short PR interval
- Less than 0.012 sec impulse originate rather than SA node

QRS Complex

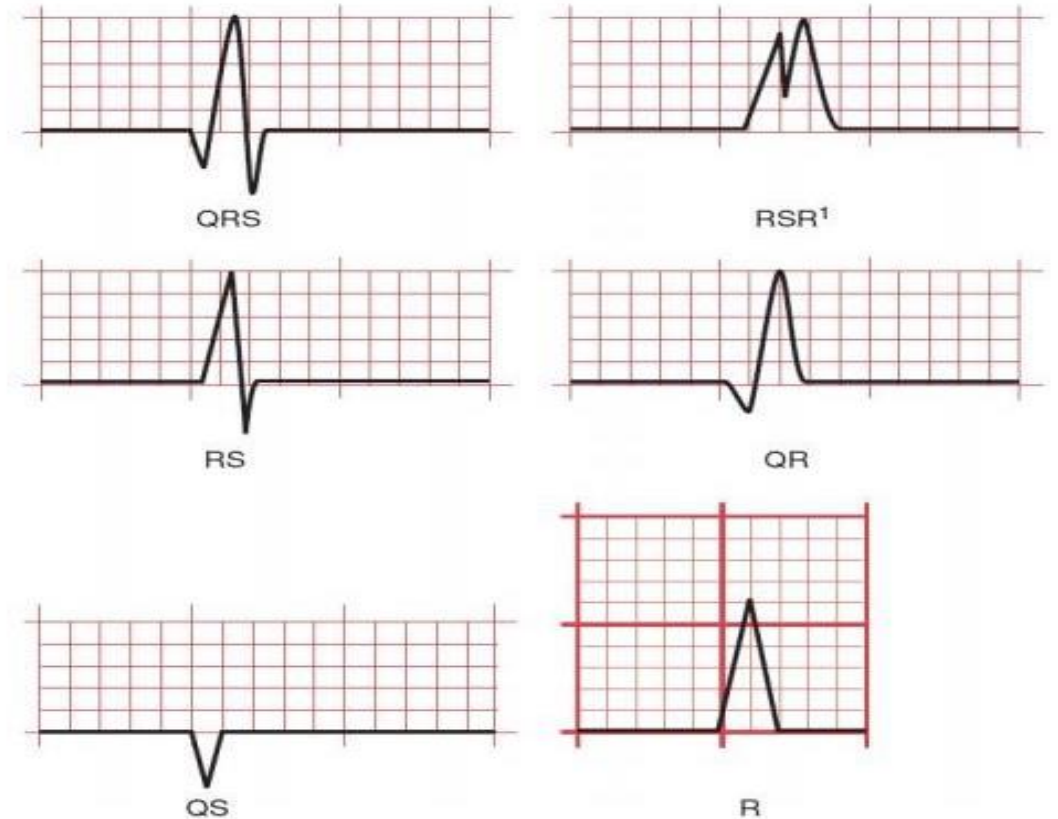
The QRS complex is a large waveform representing ventricular depolarization.

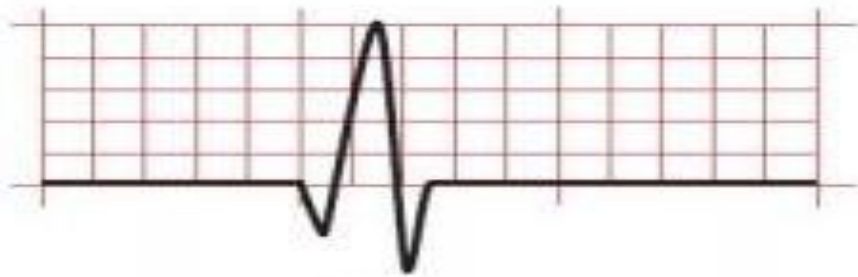
- **Location:** Follows PR interval

Configuration : The initial negative deflection is a Q wave.

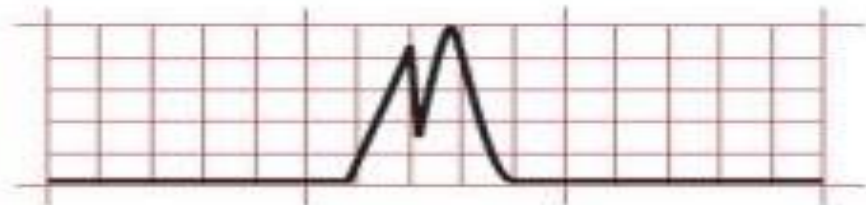
- The First positive deflection is an R wave after Q wave.
- The negative deflection after the R wave is an S wave.

Duration: A normal QRS complex duration is 0.06 to 0.10 second in width.





QRS



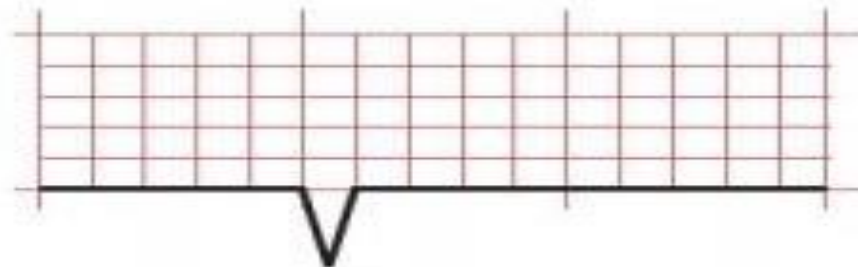
RSR¹



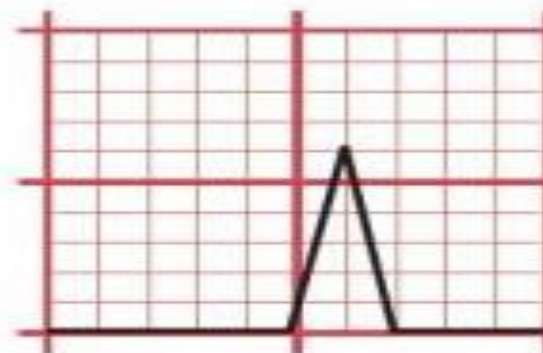
RS



QR

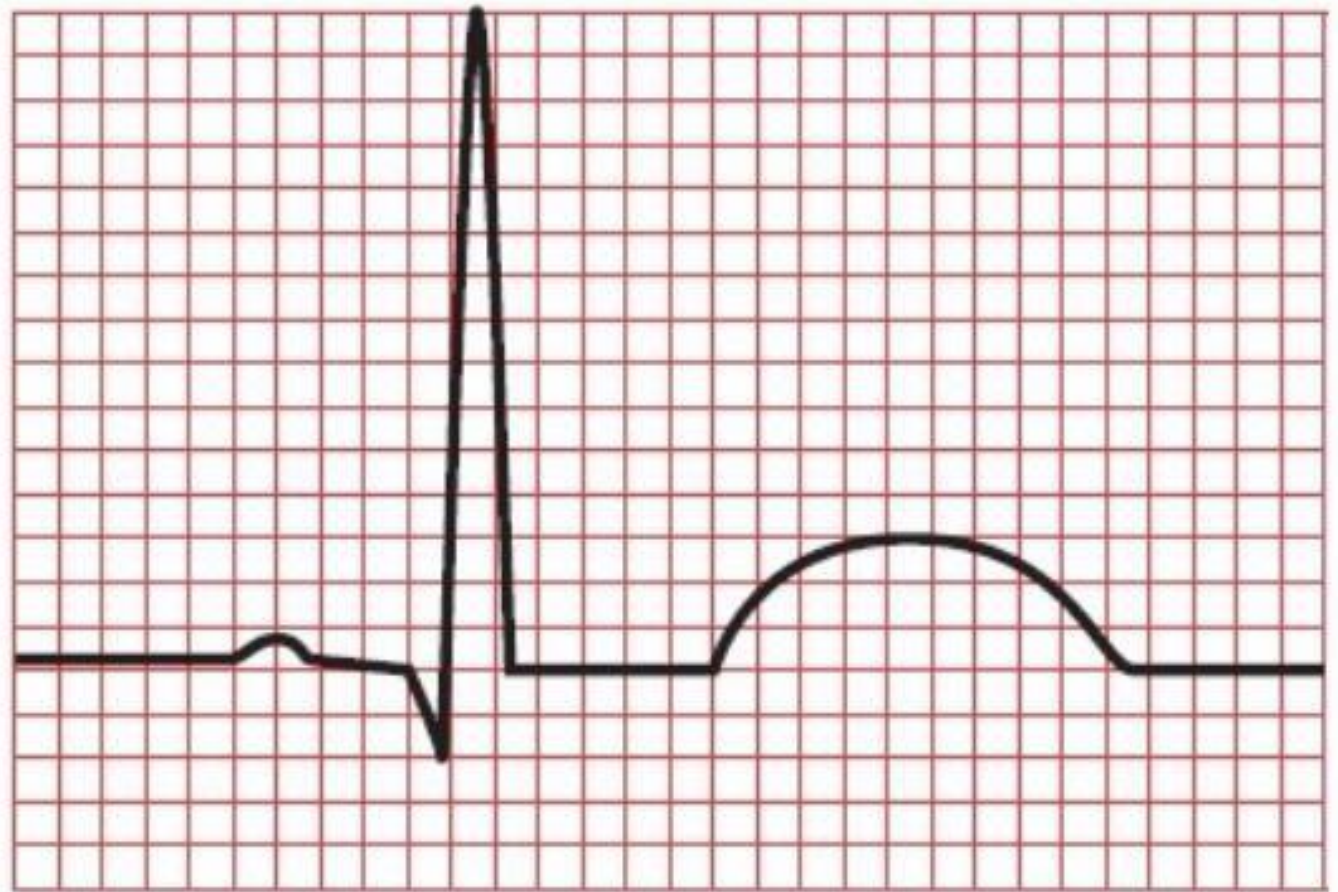


QS

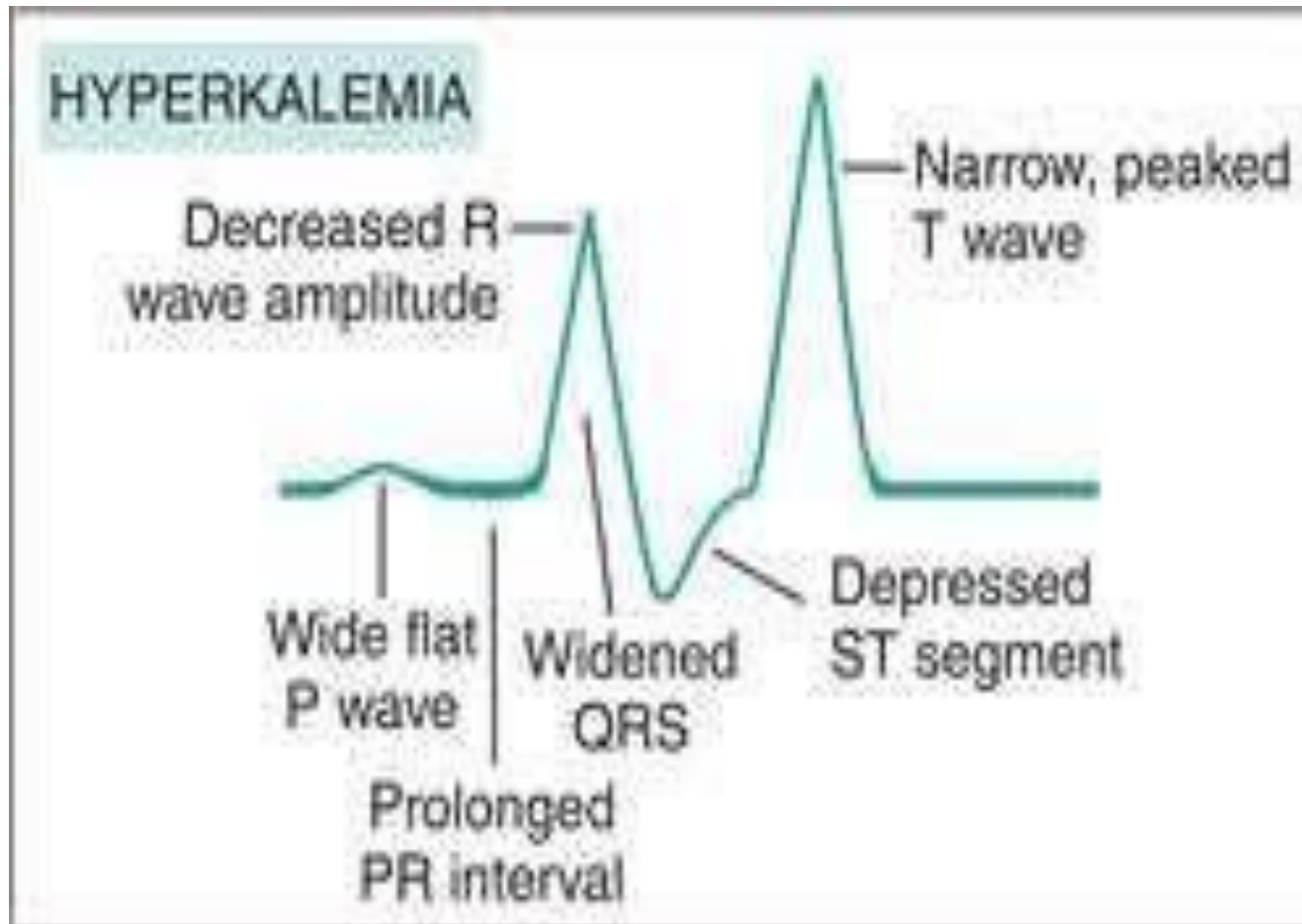


R

T wave:



T wave



- The T wave is representing ventricular repolarization or recovery.
- **Location:** The T wave appears after the QRS Complex.
- **Tall and peaked t wave**

Deflection:

- Positive or upright in lead I, II & V2 to V6.
- (N.B) the atria also have a repolarization phase.
- However, there is no visible wave on the ECG to represent atrial repolarization because it occurs at the same time as the QRS complex.

Normal



Hyperkalemia



- Tall “peaked” T wave

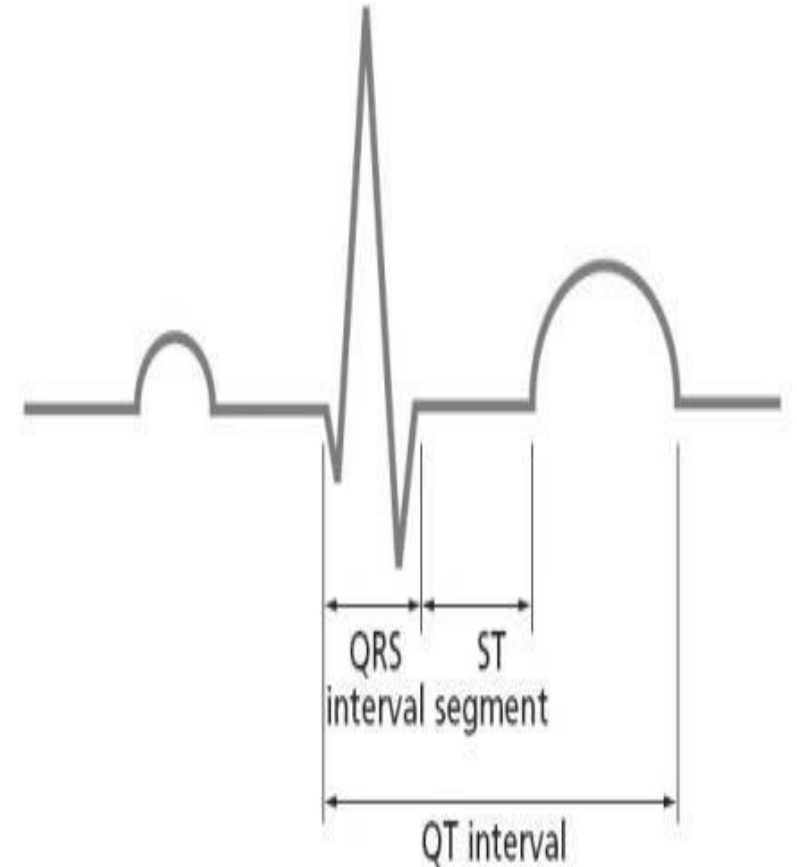
Severe
hyperkalemia

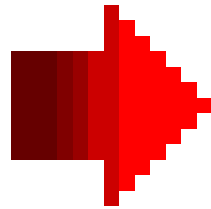


- Flattened P
- Widened QRS

ST segment

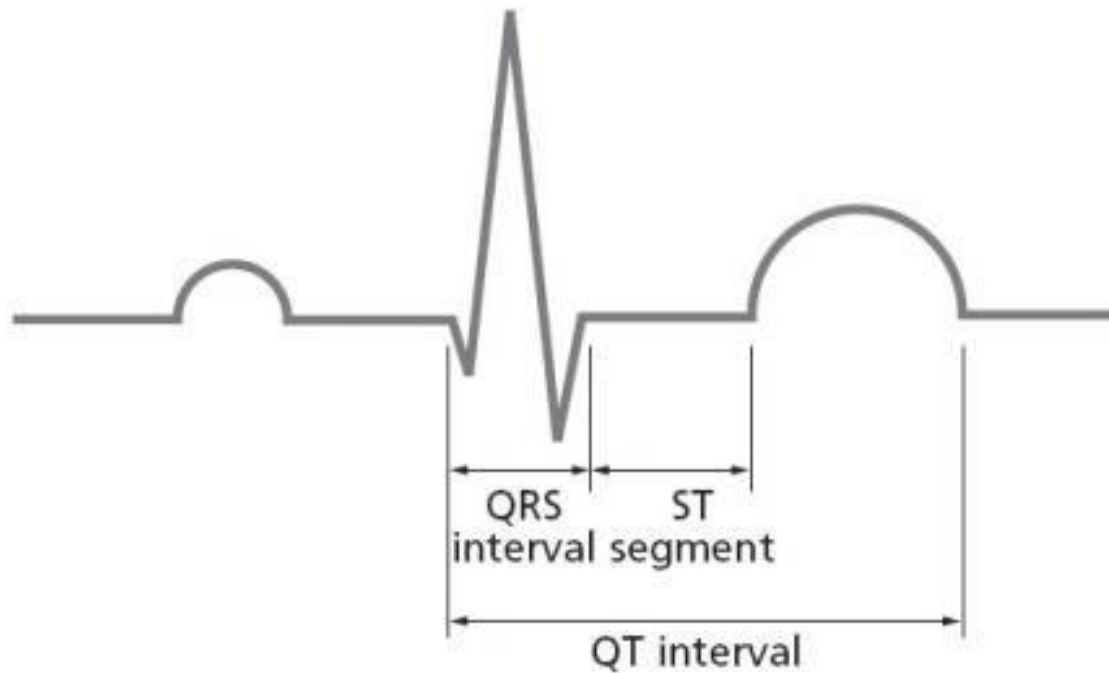
- The ST segment is the portion of the tracing from the end of the QRS complex to the beginning of the T wave.
- It represents the time from the end of ventricular depolarization to the beginning of ventricular repolarization.
- Normally, it is isoelectric.
- An isoelectric ST segment means that the ST segment joins the QRS complex at the baseline.





ST segment





-
- **QT interval.**

QT interval

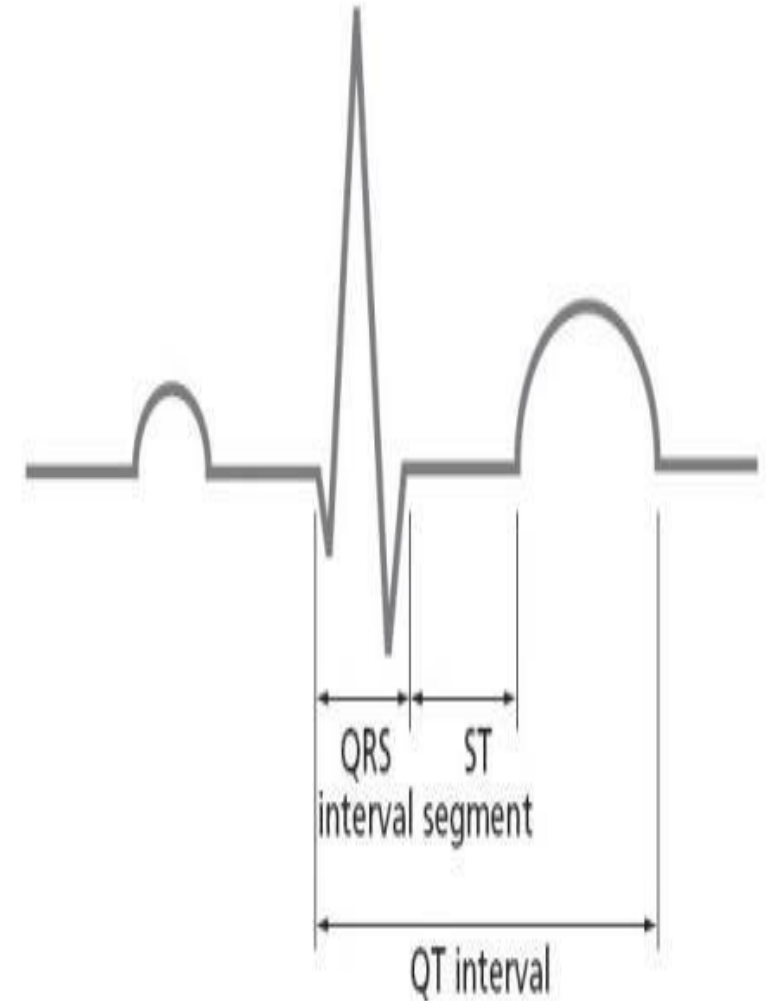
- The **QT interval** is measured from the beginning of the QRS complex to the end of the T wave and indicates the total time interval from the onset of depolarization to the completion of repolarization.
- The faster the heart rate is, the shorter the QT interval.

Location:

- extended from the beginning of QRS complex to the end of the T wave.

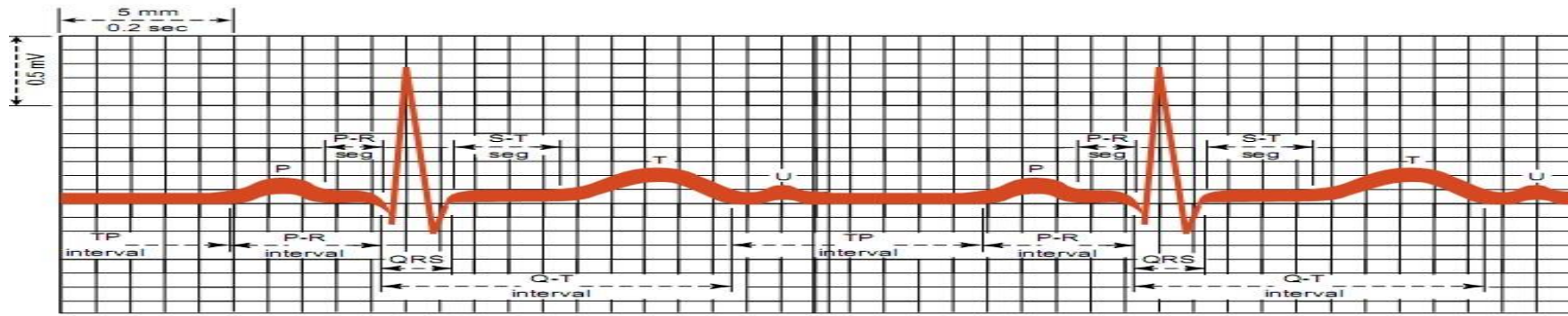
Duration:

- varies according to age, sex, and heart rate ; usually lasts from 0.36 to 0.44 second.



U wave

- **U wave** is a rarely seen, small, usually positive deflection after the T wave. The U wave represents repolarization of small segment of the ventricles.
- Large U waves may be present in patient with hypokalemia.



Analyzing a Rhythm Strip

Steps of Analyzing a Rhythm Strip

- **Determine the regularity of the rhythm**
 - Regularity is assessed by using calipers or a piece of EKG paper.
 - *To determine atrial regularity, identify the P wave and place one caliper point on the peak of the P wave.
- Located the next P wave and place the second caliper on its peak.
- If the first caliper point lands exactly on the next P wave, the atrial rhythm is perfectly regular.

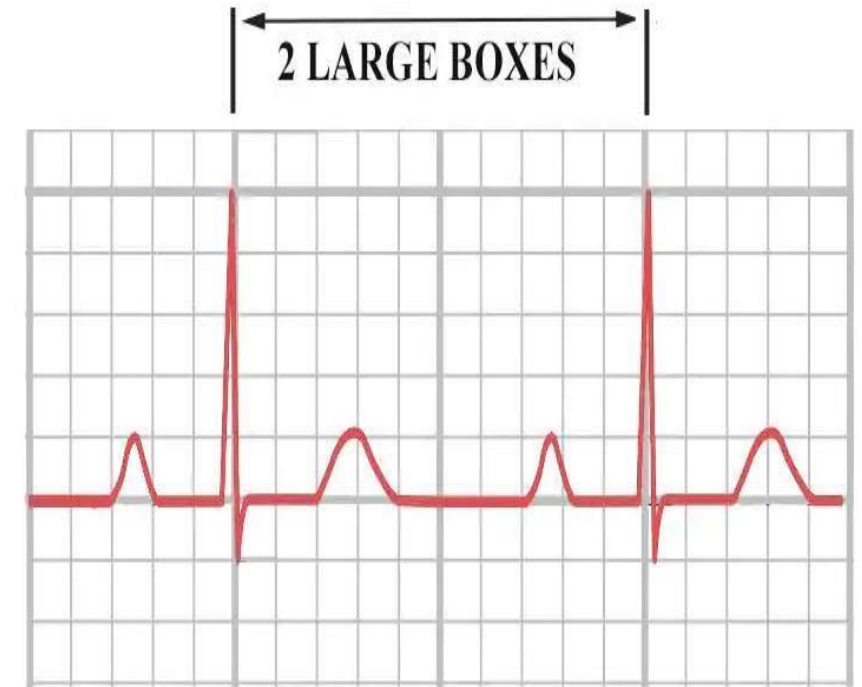
- **Determine the regularity of the rhythm**
- To determine ventricular rhythm regularity measure the intervals between two consecutive R wave, if these intervals are equal, the rhythm is consider regular.
- If the rhythm varies by 0.08 seconds (2 small squares), the rhythm would still be considered regular

Calculate the Heart Rate

- Under normal condition, the atria and ventricles depolarize in regular sequence.
- Methods of Calculating the Heart Rate
- Large Box Method
- Small Box Method
- Six –Second Method

Large Box Method: (Regular Rhythm)

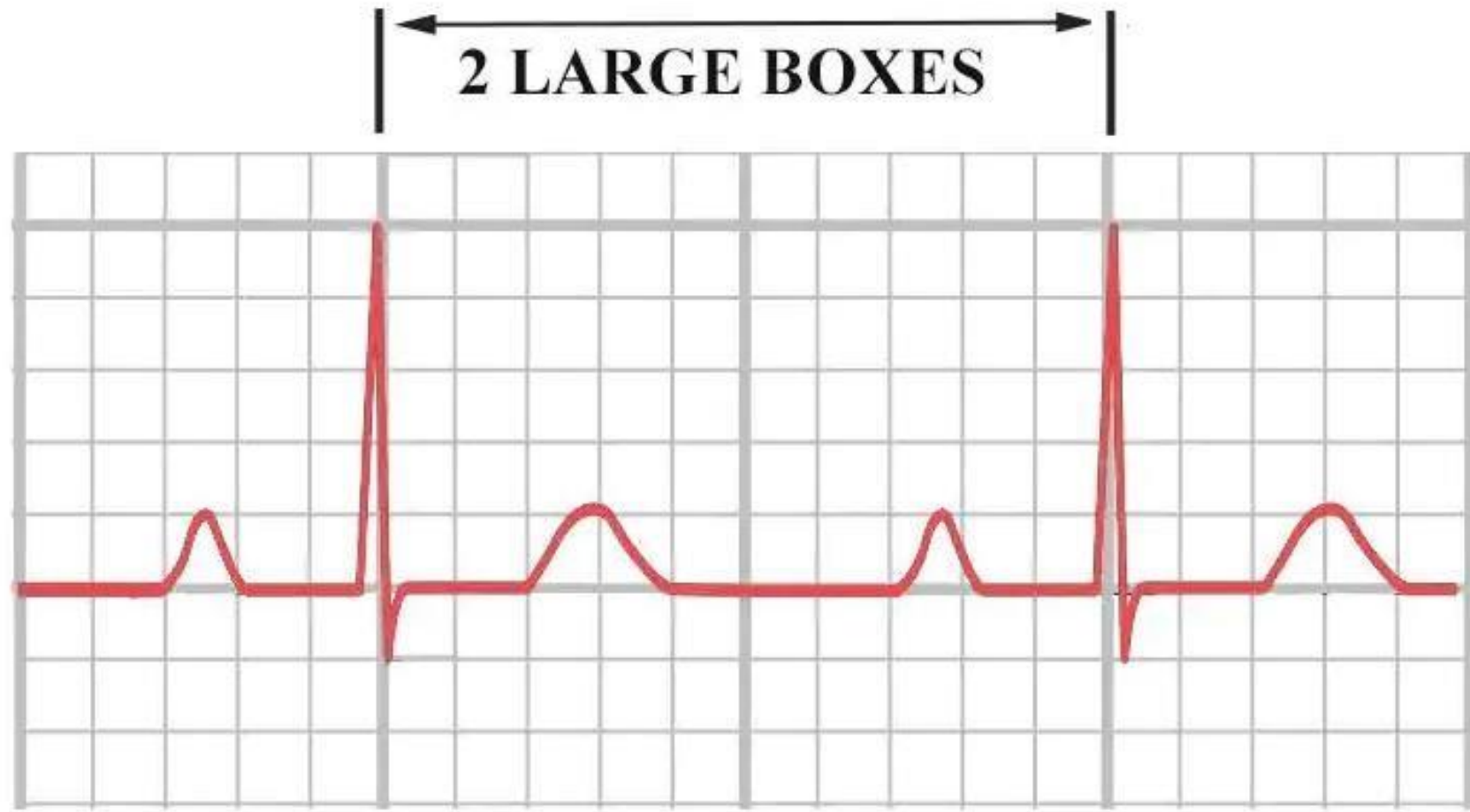
- In this method, two consecutive P waves and QRS complexes are located.
- *The number of large boxes between the highest points of consecutive P wave is counted, and the number of large boxes is divided into 300 to determine atrial rate.



Heart Rate = 300 divided by 2 large boxes = 150 bpm

Large Box Method: (Regular Rhythm)

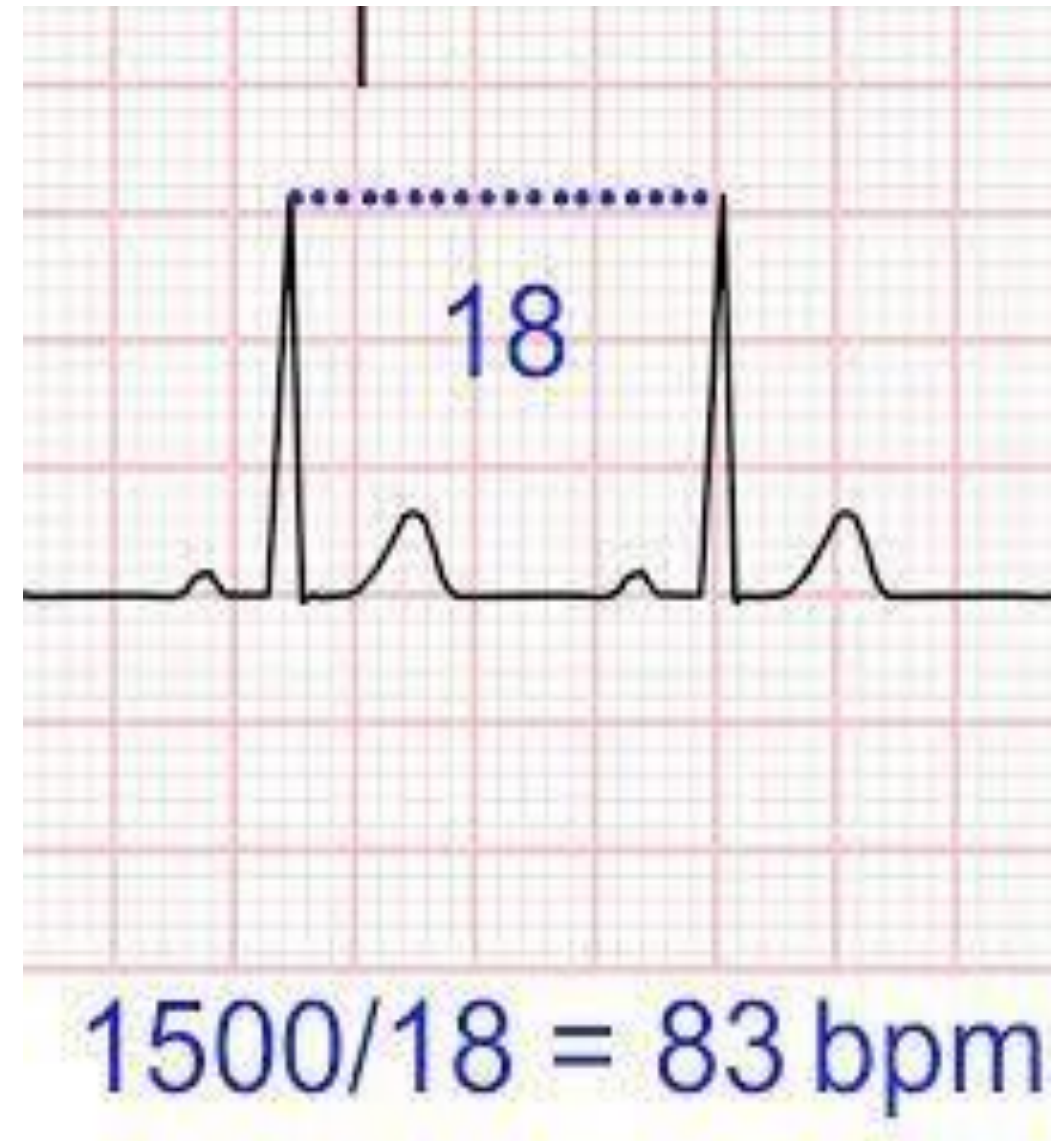
- *The number of large boxes between the highest points of consecutive QRS complexes is counted, and the number of large boxes is divided into 300 to determine ventricular rate.
- **Example:** If there are 4 large squares between regular QRS complexes, the heart rate is 75 ($300/4=75$)



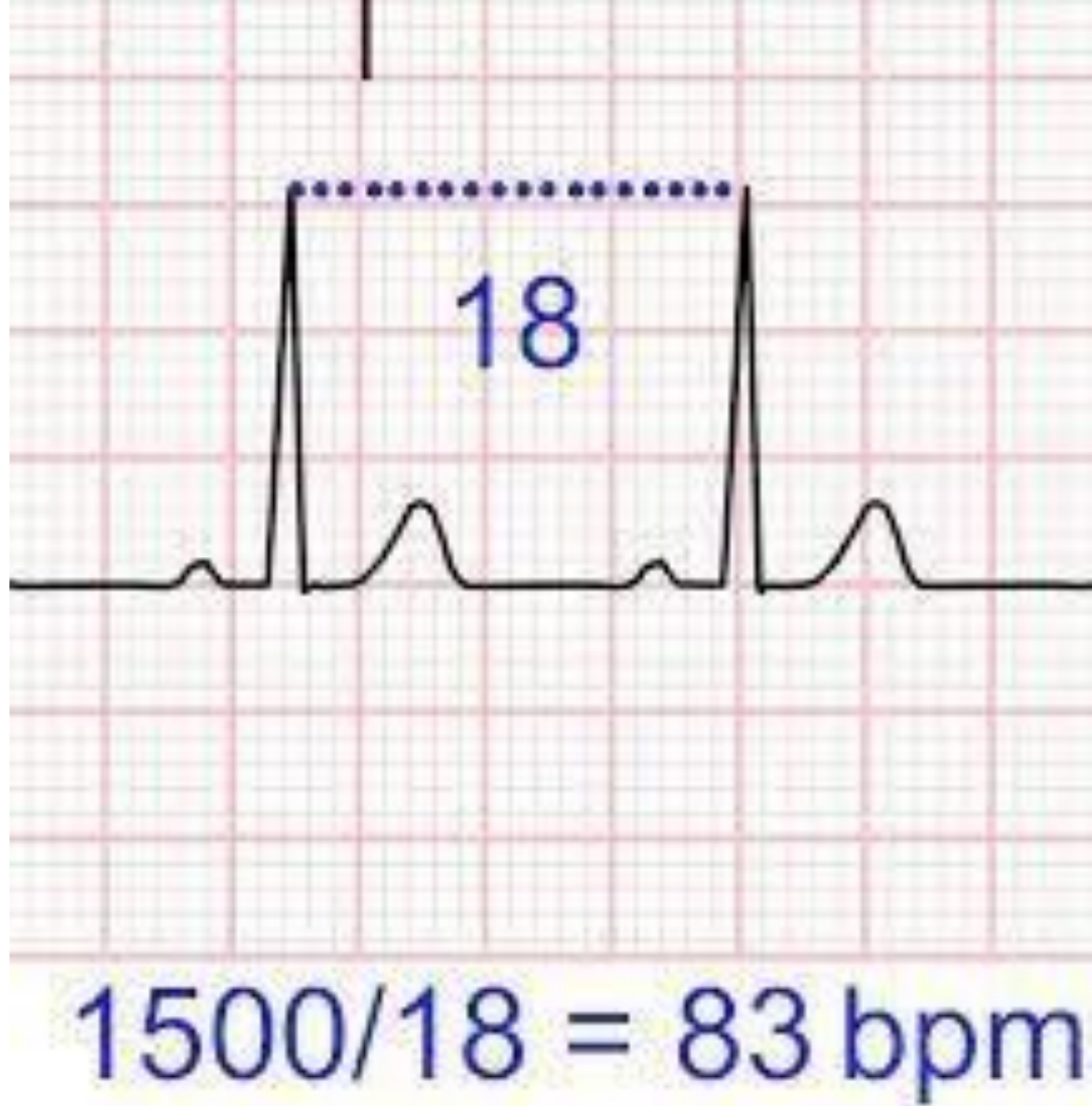
Heart Rate = 300 divided by 2 large boxes = 150 bpm

- Small Box Method: (Regular Rhythm)

- The number of **small boxes** between the highest points of consecutive P wave is counted, and the number of small boxes is divided into 1500 to determine atrial rate.
- *The number of small boxes between the highest points of consecutive QRS complexes is counted, and the number of small boxes is divided into 1500 to determine ventricular rate.



- **Small Box Method:**
(Regular Rhythm)



? Six –Second Method (Irregular Rhythms)

- Both the atrial and the ventricular rates can be estimated by examining the EGG.
- The ventricular rate is the number of QRS complexes in a 6-second rhythm strip.
- Counting the number of R waves in a 6-second strip and then multiplying that number by 10, this yields heart rate for 60 seconds or 1 minute.



The heart rate = 100

3) Evaluate P Wave

- -Are P waves present?
- -Do the P waves have a normal configuration?
- -Does the entire P wave have a similar size and shape?
- -Is there one P wave for every QRS complex?

4) Determine PR Interval Duration

- To measure the **PR interval**, count the small squares between the start of P wave and the start of QRS complex; then multiply the number of squares by 0.04 second.
- Normal PR Interval is 0.12- 0.20 seconds
- After performing this calculation ask these questions:
- Does the duration of PR interval within normal limit?
- Is the P R interval constant?

5) Determine QRS Complex Duration and shape

- Count the number of small squares between the **beginning and the end of QRS complex** & multiply this number by 0.04 second.
- -Does the duration of the QRS complex fall within normal limit?
- The normal duration of the QRS complex ranges from 0.06 to 0.10 second.
- -Are all QRS complex the same size and shape?
- -Does a QRS complex appears after every P wave?

6) Evaluate the T Wave

- -Are T waves present?
- -Do all of the T waves have a normal shape?

7) QT evaluation.

- The length of the QT interval varies with the heart rate. The QT interval is shorter when the heart rate is faster.

NORMAL SINUS RHYTHM

- The electrical impulses are formed in the SA node & discharged regularly at a rate of 60 – 100 times /minute

Criteria for identifying Normal Sinus Rhythm:

- Regular rhythm
- Both atrial & ventricular rate are the same (*Rate*: 60 to 100 beats/min)
- P waves, all the same shape, with only one preceding each QRS complex
- P waves precede each QRS complex
- The PR interval is 0.12 to 0.20 seconds
- The QRS complex duration is 0.06 to 0.12 second

post test

- Mention the conductive system of the heart
- Mention how to calculate normal rhythm of the heart rate

Any question





Thank
you